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Target Role

Senior Data Engineer

Summary

Senior Data Engineer with 10+ years of experience building scalable data platforms focused on data quality, governance, and business impact. Expertise in medallion architectures, reusable data frameworks, and data modeling to enable actionable insight and decision-making.

Proven ability to operate in ambiguous environments, define engineering standards, and treat data pipelines as products. Experience includes Palantir-powered platforms (DCGS-A CD2, Army Vantage) supporting unified, decision-ready data projects.

Core Skills

Languages: Python, SQL

Data Processing: Spark (PySpark), Pandas, dbt

Architecture: ETL/ELT, Medallion Architecture, Data Modeling

Streaming: Kafka

Data Quality: Validation Frameworks, Schema Enforcement, Anomaly Detection

Databases: PostgreSQL, NoSQL, Qdrant, pgvector

Platforms: Databricks, Kubernetes, Linux

Concepts: Data Governance, Lineage, Incremental Processing, Pipeline Abstraction

Modeling: Dimensional Modeling, Analytical Design

Orchestration: Workflow Design, Dependency Management

Professional Experience

Senior Data Engineer (Platform & Data Engineering)

12/2023 – Present

Ennoble First (SAFFIRE/CHINOOK)

- Architected and maintained scalable ETL/ELT pipelines using PySpark, enabling reliable data processing across distributed systems
- Designed and implemented a medallion architecture (Bronze/Silver/Gold) to standardize data ingestion, transformation, and consumption across analytical workflows
- Established a reusable data quality framework with rule-based validation, anomaly detection, and quarantine handling to ensure data integrity and trust in downstream datasets
- Developed modular transformation libraries to standardize business logic and accelerate development across multiple pipelines and teams
- Implemented incremental processing strategies using metadata tracking (e.g., last-processed timestamps), significantly improving pipeline efficiency and scalability
- Modeled curated datasets aligned to business and analytical use cases, enabling stakeholders to derive actionable insights from trusted data sources
- Applied data science techniques and tooling to validate transformations, characterize expected data behavior, and proactively identify anomalies in production data
- Contributed to system design and architectural decisions in an evolving environment, helping define scalable patterns and engineering standards
- Partnered with cross-functional teams to align data engineering solutions with operational needs and decision-making objectives
- Treated data pipelines as products, ensuring outputs were reliable, well-modeled, and aligned to downstream decision-making needs

Data Engineer

09/2023 – 12/2023

Los Alamos National Laboratory (Programming & Runtime Environment)

- Created and deployed RHEL and Ubuntu servers in unclassified and DOE-classified environments with custom services.
- Designed and deployed microservice architecture for license management supporting HPC utilization.
- Created, deployed, and monitored Kubernetes clusters hosting Docker containers to standardize delivery; increased team productivity by ~10%.

Software Engineer / Data Engineer

05/2021 – 12/2023

Army Software Factory (Army Futures Command)

- Built and optimized large-scale ETL pipelines processing 500M+ records, reducing runtime from 24 hours to 26 minutes and enabling faster downstream decision-making
- Developed data pipelines aggregating messaging and internal system data, integrating with enterprise analytics platforms (including Army Vantage, a Palantir-powered system)
- Enabled real-time and near-real-time visibility into vehicle fleet readiness and maintenance by delivering curated datasets into Army Vantage (Palantir-powered enterprise analytics platform)
- Ingested and normalized data from APIs and external systems into relational stores, supporting unified data views across disparate sources
- Contributed to data integration efforts aligning with enterprise “single pane of glass” initiatives, enabling cross-system visibility and improved operational planning

Data Engineer

04/2021 – 09/2022

Texas ProTax

- Aggregated disparate datasets (SQL/NoSQL) to provide curated training and analytics datasets for AI/ML workflows.
- Conducted EDA using Pandas/NumPy/scikit-learn to characterize data and inform model selection and tuning.

Digital Intelligence Systems Master Gunner (DISMG)

03/2017 – 04/2021

United States Army

- Supported implementation and operational integration of DCGS-A Capability Drop 2 (CD-2), a Palantir Gotham-backed intelligence data platform (Army Intelligence Data Platform) designed to unify and operationalize disparate data sources into a scalable data fabric
- Worked within a modernized intelligence data ecosystem enabling data ingestion, integration, visualization, and analysis across multiple sensors and systems
- Enabled data-driven decision-making by ensuring reliability, accuracy, and timeliness of intelligence data systems in operational environments
- Performed data validation, system troubleshooting, and data flow optimization to maintain integrity and availability of mission-critical data
- Developed deep understanding of large-scale data integration, data architecture, and how data latency and quality impact real-time analytical outcomes

Intelligence Analyst / Data Analyst → Data Scientist → Data Architect 09/2002 – 04/2024 United States Army

- Progressed across roles from analyst to architect, developing end-to-end understanding of data lifecycle from ingestion and modeling to advanced analytics and decision support
- Applied data analysis techniques to multi-source intelligence datasets, performing data cleansing, correlation, and trend analysis.
- Built statistical and predictive models to support forecasting and decision-making under uncertainty.
- Designed and maintained analytical data structures and pipelines to support repeatable analysis at scale.
- Served as a technical bridge between analysts and engineers, defining data requirements and system architecture.

Selected Platform & Data Engineering Projects

Classification-Aware Federated Pareto Optimization

Python, PyTorch, Federated Learning

- *Architecture:* Distributed federated learning system with multi-objective optimization across heterogeneous clients
- Designed a classification-aware federated learning approach to identify Pareto-efficient tradeoffs across performance, fairness, and resource cost
- Implemented multi-objective optimization to support decision-making across constrained and distributed environments
- Evaluated convergence and tradeoff stability versus baseline federated learning strategies

Large-Scale Medallion Data Pipeline

PySpark, Delta Lake, Parquet

- *Architecture:* Medallion (Bronze/Silver/Gold) batch processing pipeline with integrated data quality and incremental processing
- Architected an end-to-end medallion pipeline for large analytical datasets using PySpark
- Implemented data quality rules and investigation tables to isolate invalid records while preserving analyst visibility
- Optimized transformations and partitioning to support scalable downstream analytics and ML workloads

Geospatial Analytics and Visualization Platform (DoD / SAFFIRE Context) *Databricks, Apache Sedona, H3, AWS*

- *Architecture:* Distributed geospatial data platform leveraging medallion modeling and spatial indexing (H3)
- Built a geospatial analytics pipeline backed by medallion tables to support analyst workflows at scale
- Integrated H3 indexing and WKT geometries to enable performant spatial joins, clustering, and filtering
- Delivered interactive map-based analysis using geospatial SQL and Kepler.gl for operational use cases

Kubernetes-Based Local Data Science Environment *Kubernetes, Docker, Postgres, Qdrant/pgvector*

- *Architecture:* Containerized, service-oriented data platform with relational and vector storage integration
- Designed a reproducible Kubernetes environment for data engineering and ML experimentation aligned to cloud-native patterns
- Integrated relational and vector databases to support retrieval, embedding, and hybrid query workflows
- Improved developer productivity through scripted orchestration, service composition, and observability-friendly configuration

Data Modeling and Validation with dbt *dbt, SQL*

- *Architecture:* Modular transformation layer with test-driven data modeling and dependency-managed workflows
- Designed and implemented modular data models using dbt, including schema tests and data quality validation
- Explored orchestration patterns and dependency management for scalable, reproducible transformations

End-to-End Data Pipeline (Personal Lab) *Kafka, Spark, Python, Kubernetes*

- *Architecture:* Hybrid streaming + batch pipeline with event ingestion, processing, and modular transformation layers
- Built streaming and batch pipelines using Kafka and Spark in a Kubernetes-based environment
- Integrated data quality validation, incremental processing, and modular architecture patterns

Vector Search and Data Platform Exploration *Qdrant, pgvector, RAG*

- *Architecture:* Hybrid retrieval system combining vector similarity search with structured data storage
- Experimented with vector databases and RAG-style pipelines to understand modern data + AI system design

Education & Certifications

- M.S. Data Engineering, Auburn University (Expected 2027)
- B.S. Database Management / Data Analytics, Western Governors University
- ITILv4 — CIW-Web Foundations — CIW-Data Analyst — CompTIA Project+